

§11.6 Absolute Convergence and The Ratio and Root Tests

Homework : 1,7,10,11,15,18,19,23,27,31,32,33,40

Definition :

- i. $\sum_{n=1}^{\infty} a_n$ is said to be absolute convergence (AC)
if $\sum_{n=1}^{\infty} |a_n|$ is convergent.
- ii. If $\sum_{n=1}^{\infty} a_n$ is convergent but not AC, then $\sum_{n=1}^{\infty} a_n$ is called conditional convergence(CC).

Example 1 :

- i. $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$: CC
- ii. $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$: AC

Theorem1 : AC $\Rightarrow$ convergence (反之不對).

Ratio and Root Tests (for AC)

Who : anyone

What : 這 2 個方法的精神都是在檢查尾巴項的行為。且都以等比級數 $\sum_{n=1}^{\infty} a_0 r^n$ 作為比較的對象。

(i) Ratio Test : 比前後項縮小比例，即

- $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ (若為等比級數， L 即為公比。)
- 適用對象有 ! 項.

(ii) Root Test : • $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$

(若為等比級數，則 $\sqrt[n]{|a_0| r^n} = |a_0|^{\frac{1}{n}} r = r$, 即 L 也為 r ,

即開 n th 根號也為檢查尾巴項行為的方法之一).

- 適用對象有 $(\quad)^n$ 項.

結論：

- (i) $L < 1 \Rightarrow$ AC
- (ii) $L > 1 \Rightarrow$ Divergence
- (iii) $L = 1 \Rightarrow$ The test fails. $\left(\sum_{n=1}^{\infty} \frac{1}{n}, \sum_{n=1}^{\infty} \frac{1}{n^2} \right)$
(P-series 用此 2 tests, 2 都檢查不出來。)

註： $n^n \gg n! \gg e^n \gg n^2 \gg \ln$.

不同量級，重量級在分母，必 AC.

Example 2 : Determine whether the series is AC, CC or divergent.

	Test	AC, CC or Divergent
i. $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$	$\sqrt{\quad}, \frac{n^n}{2} = \frac{1}{2}$	AC
ii. $\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+1} \right)^n$	$\sqrt{\quad}, \frac{1}{2}$	AC
iii. $\sum_{n=1}^{\infty} \frac{(-3)^n}{n!}$	$\text{---}, \left \frac{3}{n+1} \right \rightarrow 0$	AC
iv. $\sum_{n=1}^{\infty} (-1)^n \frac{2^n n!}{5.8 \dots (3n+2)}$	$\text{---}, \left \frac{2(n+1)}{3(n+1)+2} \right \rightarrow \frac{2}{3}$	AC
	Div.	
v. $\sum_{n=1}^{\infty} (-1)^n \frac{5^n n!}{5.8 \dots (3n+2)}$	$\text{---}, \left \frac{5(n+1)}{3(n+1)+2} \right \rightarrow \frac{5}{3}$	Divergent
vi. $\sum_{n=1}^{\infty} \frac{(-1)^n}{(\tan^{-1} n)^n}$	$\sqrt{\quad}, \left \frac{1}{\tan^{-1} n} \right \rightarrow \frac{2}{\pi}$	AC
vii. $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n\sqrt{n}}$	$\approx \sum_{n=1}^{\infty} \frac{\frac{\pi}{2}}{n^{\frac{3}{2}}} \Rightarrow$	AC

註：P-級數 不需用（也不能用）此 2 tests.

Example 3 : For which positive integers k is the following series convergent?

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}$$

Solution :

$$\frac{\frac{((n+1)!)^2}{(k(n+1))!}}{\frac{(n!)^2}{(kn)!}} = \frac{(n+1)^2}{(kn+1)(kn+2)\dots(kn+k)}$$

$\Rightarrow$ 當 $k \geq 2$, $L < 1$. (即收斂)

補充 :

Sterling Formula.

$$\left(\frac{n}{e}\right)^n \sqrt{2\pi n} < n! < \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \left(1 + \frac{1}{4n}\right).$$

or

$$n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n}$$

Example 1 : Determine the convergence / divergence of the series

$$\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} 4^n = \sum_{n=0}^{\infty} a_n.$$

Solution : Using sterling formula, we have

$$a_n = \frac{(n!)^2}{(2n)!} 4^n \sim \frac{\left[\left(\frac{n}{e}\right)^n \sqrt{2\pi n}\right]^2}{\left(\frac{2n}{e}\right)^{2n} \sqrt{2\pi(2n)}} = \frac{2\pi n}{\sqrt{4n\pi}}$$

$$\nrightarrow 0 \quad \text{or} \quad n \rightarrow \infty$$

$\Rightarrow$ The series diverges

Example 2 : $1.3.5 \cdots (2n+1) = \frac{(2n+1)!}{2.4.6 \cdots 2n} = \frac{(2n+1)!}{2^n n!}$